INCH-POUND

MIL-DTL-32546/2A(AS)
11 July 2018
SUPERSEDING
MIL-DTL-32546/2(AS)
14 March 2016

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, PLUG, HIGH-SPEED DATA

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32546.

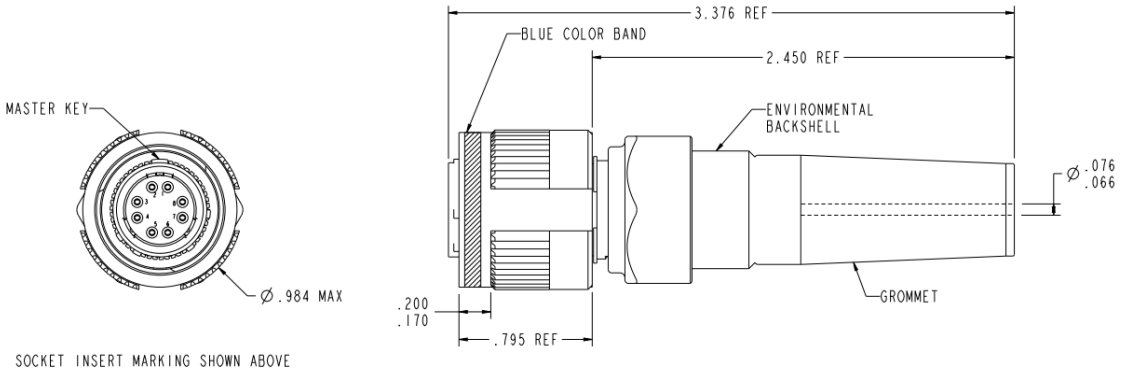


FIGURE 1. Plug connector.

TABLE I. Insert arrangement for size 11 shell.

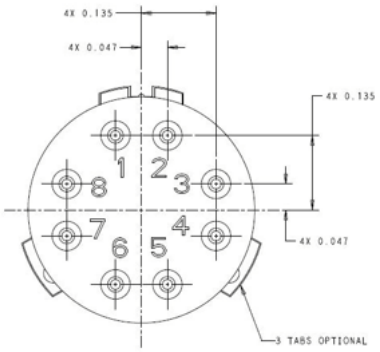
Insert Arrangement (see figure 2)	Service Rating (DWV*)
11-08	1300V rms**

*dielectric withstanding voltage
**root mean square

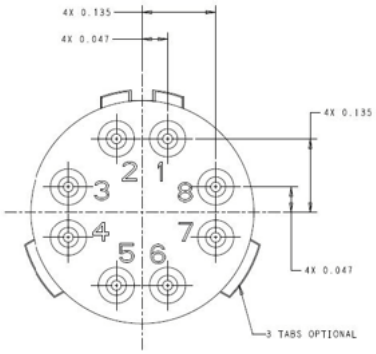
AMSC N/A

FSC 5935

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Front Face – Pin Contacts



Front Face – Socket Contacts

TABLE II. Contact location Identification/cable wire color.

Pin/ Socket 1/	Cable wire colors 2/	EIA/TIA 586A/586B Pairs 3/
1	White/Blue	1
2	Blue	
3	White/Orange	2
4	Orange	
5	White/Green	3
6	Green	
7	White/Brown	4
8	Brown	

1/ Contact location identification is for reference only (see 3.40).
2/ Recommended cable wire color locations.
3/ Recommended EIA/TIA 586 pair allocation.

FIGURE 2. 11-08 insert detail.

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TABLE III. Contact information.

PIN					
Type	SAE AS39029 Part number	SAE AS22520		SAE AS81969	
		Crimping tool 1/	Positioner 1/	Installing tool 1/	Removal tool 1/
22D Pin	M39029/58-360	M22520/2-01 M22520/7-01	M22520/2-09 M22520/7-07	M81969/8-01 M81969/14-01	M81969/8-02 M81969/14-01
	M39029/107-620				
	M39029/122-671				
Socket					
22D Socket	M39029/56-348	M22520/2-01 M22520/7-01	M22520/2-07 M22520/7-05	M81969/8-01 M81969/14-01	M81969/8-02 M81969/14-01
	M39029/106-614				
	M39029/121-664				

1/ See SAE AS39029/56, /58, /106, /107, /121 and /122 for additional crimp tool and installation/removal tool information.

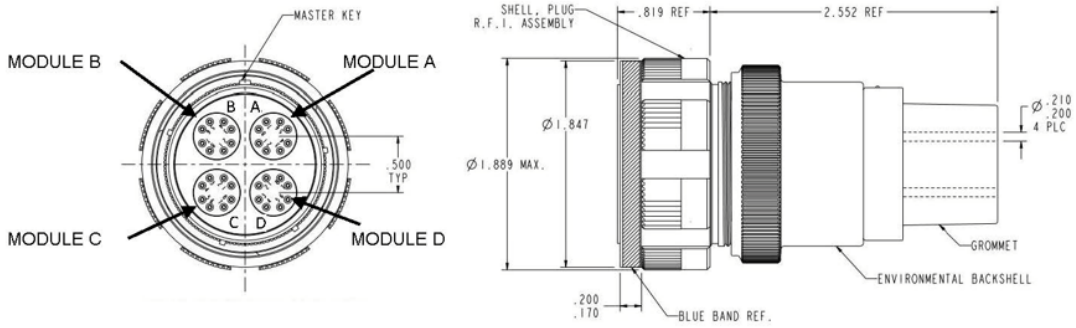


FIGURE 3. Plug connector, size 25 (pin insert arrangement example).

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TABLE IV. Insert layouts for size 25 shell.

Shell size	Arrangement no.	Number of contacts	Size contacts	Service rating	Contact location	Service rating (DWV)
25	4 X 11-08	32	22	M	All	1300V rms

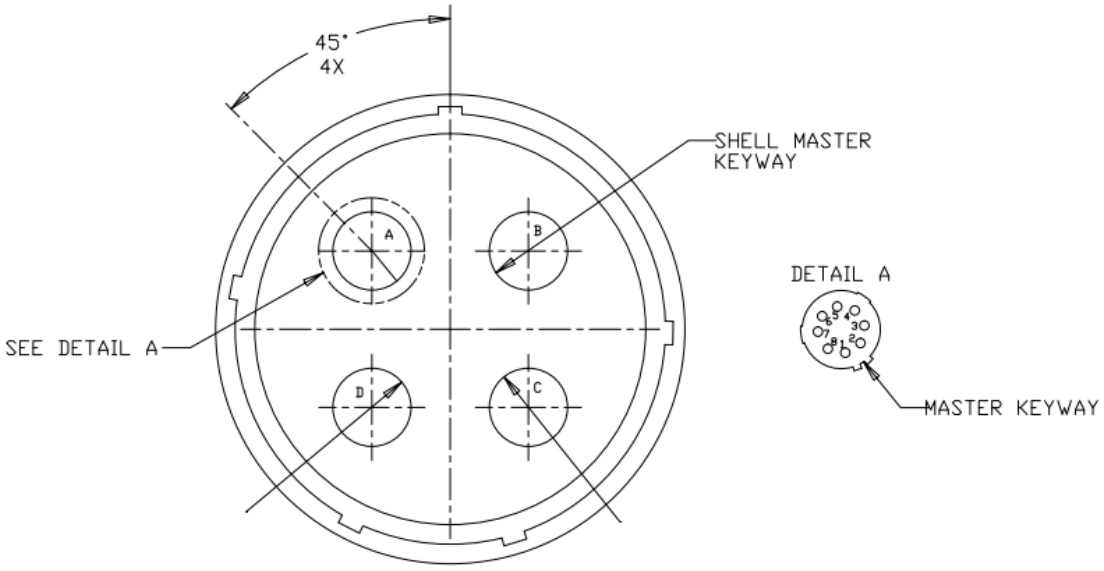


FIGURE 4. Insert orientation for size 25 shell. 1/

1/ Note that the master key of insert shall be oriented at a 135 degree angle toward the center of the connector. Insert arrangement 11-08 shown in cavity A as reference.

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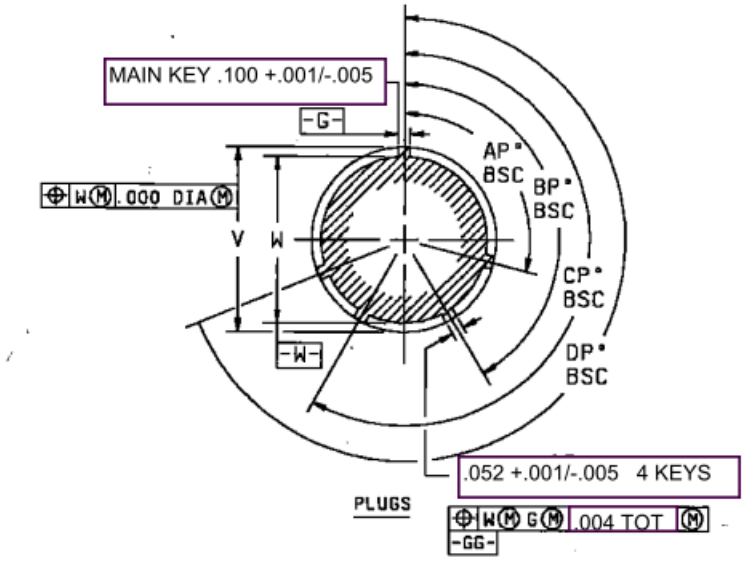


FIGURE 5. Plug polarization.

TABLE V. Main key and keyway polarization.

Shell size	Key and keyway polarization identification letter	AR° or AP° BSC	BR° or BP° BSC	CR° or CP° BSC	DR° or DP° BSC
11	N	95	141	208	236
	A	113	156	182	292
	B	90	145	195	252
	C	53	156	220	255
	D	119	146	176	298
	E	51	141	184	242
25	N	80	142	196	293
	A	135	170	200	310
	B	49	169	200	244
	C	66	140	200	257
	D	62	145	180	280
	E	79	153	197	272

NOTES:

1. Dimensions are in inches.
2. Environmental backshell included with connector kit.
3. Mating face of pin insert shown on insert layouts.

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TABLE VI. Plug with pin contact locations.

(Insert arrangement 4X 11-08)		
Contact position identification (ID)	Contact location	
	X-axis (mm)	Y-axis (mm)
A1	+.1878 (4.77)	+.1213 (3.08)
A2	+.1213 (3.08)	+.1878 (4.77)
A3	+.1213 (3.08)	+.3122 (7.93)
A4	+.1878 (4.77)	+.3787 (9.62)
A5	+.3122 (7.93)	+.3787 (9.62)
A6	+.3787 (9.62)	+.3122 (7.93)
A7	+.3787 (9.62)	+.1878 (4.77)
A8	+.3122 (7.93)	+.1213 (3.08)
B1	-.1213 (3.08)	+.1878 (4.77)
B2	-.1878 (4.77)	+.1213 (3.08)
B3	-.3122 (7.93)	+.1213 (3.08)
B4	-.3787 (9.62)	+.1878 (4.77)
B5	-.3787 (9.62)	+.3122 (7.93)
B6	-.3122 (7.93)	+.3787 (9.62)
B7	-.1878 (4.77)	+.3787 (9.62)
B8	-.1213 (3.08)	+.3122 (7.93)
C1	-.1878 (4.77)	-.1213 (3.08)
C2	-.1213 (3.08)	-.1878 (4.77)
C3	-.1213 (3.08)	-.3122 (7.93)
C4	-.1878 (4.77)	-.3787 (9.62)
C5	-.3122 (7.93)	-.3787 (9.62)
C6	-.3787 (9.62)	-.3122 (7.93)
C7	-.3787 (9.62)	-.1878 (4.77)
C8	-.3122 (7.93)	-.1213 (3.08)
D1	+.1213 (3.08)	-.1878 (4.77)
D2	+.1878 (4.77)	-.1213 (3.08)
D3	+.3122 (7.93)	-.1213 (3.08)
D4	+.3787 (9.62)	-.1878 (4.77)
D5	+.3787 (9.62)	-.3122 (7.93)
D6	+.3122 (7.93)	-.3787 (9.62)
D7	+.1878 (4.77)	-.3787 (9.62)
D8	+.1213 (3.08)	-.3122 (7.93)

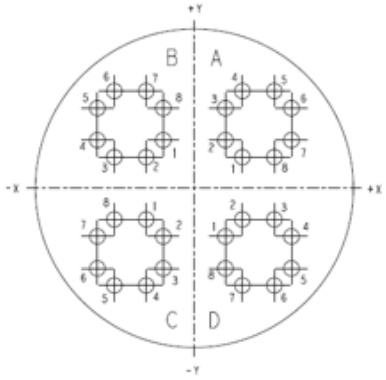


FIGURE 6. Plug with pin contacts.

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TABLE VII. Plug with socket contact locations.

(Insert arrangement 4X 11-08)		
Plug with Socket Contacts		
Contact position identification (ID)	Contact location	
	X-axis (mm)	Y-axis (mm)
A1	+.1213 (3.08)	+.1878 (4.77)
A2	+.1878 (4.77)	+.1213 (3.08)
A3	+.3122 (7.93)	+.1213 (3.08)
A4	+.3787 (9.62)	+.1878 (4.77)
A5	+.3787 (9.62)	+.3122 (7.93)
A6	+.3122 (7.93)	+.3787 (9.62)
A7	+.1878 (4.77)	+.3787 (9.62)
A8	+.1213 (3.08)	+.3122 (7.93)
B1	-.1878 (4.77)	+.1213 (3.08)
B2	-.1213 (3.08)	+.1878 (4.77)
B3	-.1213 (3.08)	+.3122 (7.93)
B4	-.1878 (4.77)	+.3787 (9.62)
B5	-.3122 (7.93)	+.3787 (9.62)
B6	-.3787 (9.62)	+.3122 (7.93)
B7	-.3787 (9.62)	+.1878 (4.77)
B8	-.3122 (7.93)	+.1213 (3.08)
C1	-.1213 (3.08)	-.1878 (4.77)
C2	-.1878 (4.77)	-.1213 (3.08)
C3	-.3122 (7.93)	-.1213 (3.08)
C4	-.3787 (9.62)	-.1878 (4.77)
C5	-.3787 (9.62)	-.3122 (7.93)
C6	-.3122 (7.93)	-.3787 (9.62)
C7	-.1878 (4.77)	-.3787 (9.62)
C8	-.1213 (3.08)	-.3122 (7.93)
D1	+.1878 (4.77)	-.1213 (3.08)
D2	+.1213 (3.08)	-.1878 (4.77)
D3	+.1213 (3.08)	-.3122 (7.93)
D4	+.1878 (4.77)	-.3787 (9.62)
D5	+.3122 (7.93)	-.3787 (9.62)
D6	+.3787 (9.62)	-.3122 (7.93)
D7	+.3787 (9.62)	-.1878 (4.77)
D8	+.3122 (7.93)	-.1213 (3.08)

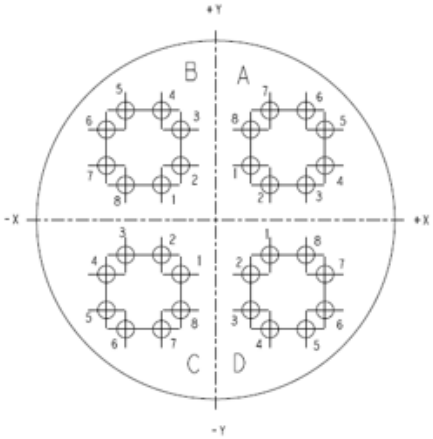


FIGURE 7. Plug with socket contacts.

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REQUIREMENTS:

1. Materials:

Resilient inserts/backshell grommet – silicone elastomer per ASTM D6411/6411M.

Plastic inserts – thermoplastic per ASTM D5205.

2. Dimensions and configuration: See Figure 1. Interface dimensions shall be in accordance with connector intermateability dimensions (series III) of MIL-DTL-38999.

3. This connector mates with MIL-DTL-32546/1 (square flange receptacle) or /3 (jam nut receptacle).

4. Part or Identifying Number (PIN) example:

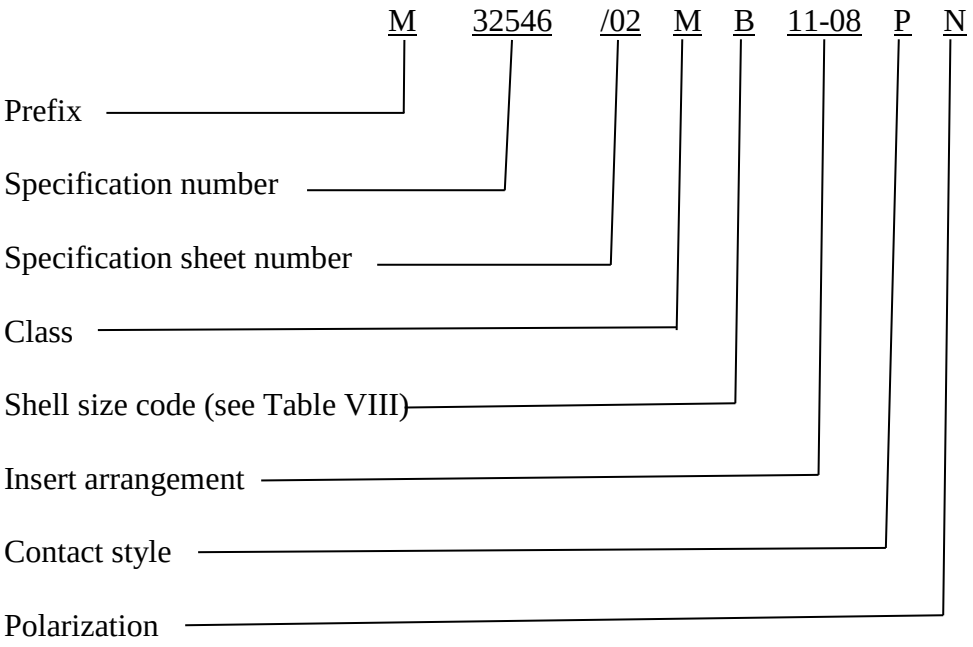


TABLE VIII. Shell size code.

Shell size	Shell size code
11	B
25	F

5. See Table III for contact information for Figure 1 and 3 plug contacts.

6. The frequency ranges is 1-500 MHz.

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7. Impedance is determined by the electrical performance and characteristics stated in MIL-DTL-32546.

8. Connector physical requirements shall meet the requirements of MIL-DTL-38999, Series III, unless otherwise noted.

9. Backshell in accordance with SAE AS85049 not applicable (see 3.5.3). The backshell shall be in accordance with Figures 1 and 3.

10. Size 25 connector shall have removable modules (see Figure 3). Cavity fills per paragraph 3.5.10 around individual modules does not apply to the individual modules of the size 25 connector.

EXAMINATION AND TESTS:

1. Additional testing.

2. Vibration for shell sizes 11 and 25 shall be conducted with the backshell and grommet shown on Figures 1 and 3, respectively. Accelerometer location (see MIL-DTL-32546, figure 2) shall be as defined in Table X. Cable weight between rear of grommet and tie down clamp point to be considered a portion of the backshell assembly weight.

3. Group 3 and Group 7 testing, Class M and J composite connectors may be tested with aluminum backshells. With aluminum backshells the class M connector will be rated at the 48 hour salt spray requirement instead of the 500 hour salt spray requirement.

4. High impact shock shall be tested with the backshell shown on Figures 1 and 3.

5. Group 4 testing shall be conducted using shell size 25.

6. Group 11 testing shall be conducted using shell sizes 11 and 25.

7. The backshell shown on Figures 1 and 3 shall terminate to the braid of category 6A overall shielded cable. This backshell is not intended to be used with a braidsock. Lightning strike tests are not applicable in group 11 testing.

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TABLE IX. Cable braid retention test.

Inspection	Requirement	Test Paragraph
Cable braid retention	12.5 pounds force with slippage not to exceed 0.125 in.	SAE AS85049 para. 4.6.9 (Method A ECIA EIA-364-38) except that the pull force is to be applied to the cable braid (less jacket).1/

1/ In the event of multiple cables, test applies to individual cables.

TABLE X. Accelerometer locations.

Shell size	Weight (kg) +10%/-0%	LL (mm) ± 1
11	.035	10.8
25	.13	19.2

NOTES:

1. Intended use. The connectors qualified to this specification are intended for use as high speed data connectors using copper conductor for transmission of signals up to 10 gigabits per second (Gbps) such as IEEE 802.3 10GBASE-T, over four-pair cabling.

Referenced documents. In addition to MIL-DTL-32546 and its applicable specification sheets, this specification sheet references:

ECIA EIA-364-38	SAE AS22520	SAE AS39029/106
EIA/TIA 586	MIL-DTL-32546/1	SAE AS39029/107
IEEE 802.3	MIL-DTL-38999	SAE AS39029/121
ASTM D5205	SAE AS39029/56	SAE AS39029/122
ASTM D6411/D6411M	SAE AS39029/58	SAE AS81969
SAE AS85049		

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Preparing activity:
Navy – AS
Project 5935-2018-033

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